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ShredTracker Database Re-Design

Changes from old design

The six MVP tables will now use singular names: User, Workout, WorkoutSet, Exercise, WeightEntry and WeightGoal. Their primary keys and fields will remain the same. Saved workout routines are added as a stretch feature for after the MVP is completed and deployed.

Database Choice

My database choice for ShredTracker will be MySQL, a relational database. The application will store the following data: users, workouts, exercise sets, body weight entries and weight loss goals. One user can have multiple workouts, and each workout can contain multiple sets. MySQL will store the original records, while the service layer will calculate trends and progress from those records.

How the application uses the data

A user can log their sets and compare the performance with an earlier workout. The application will calculate volume by adding weight multiplied by reps for each set of the exercise. The application will also keep record of their highest personal record. A user can also enter their body weight and set a weight loss goal. The application will use their data entry to show a line chart of progress towards their targeted goal. Users can also review their workout history to see their previous exercise and what was completed.

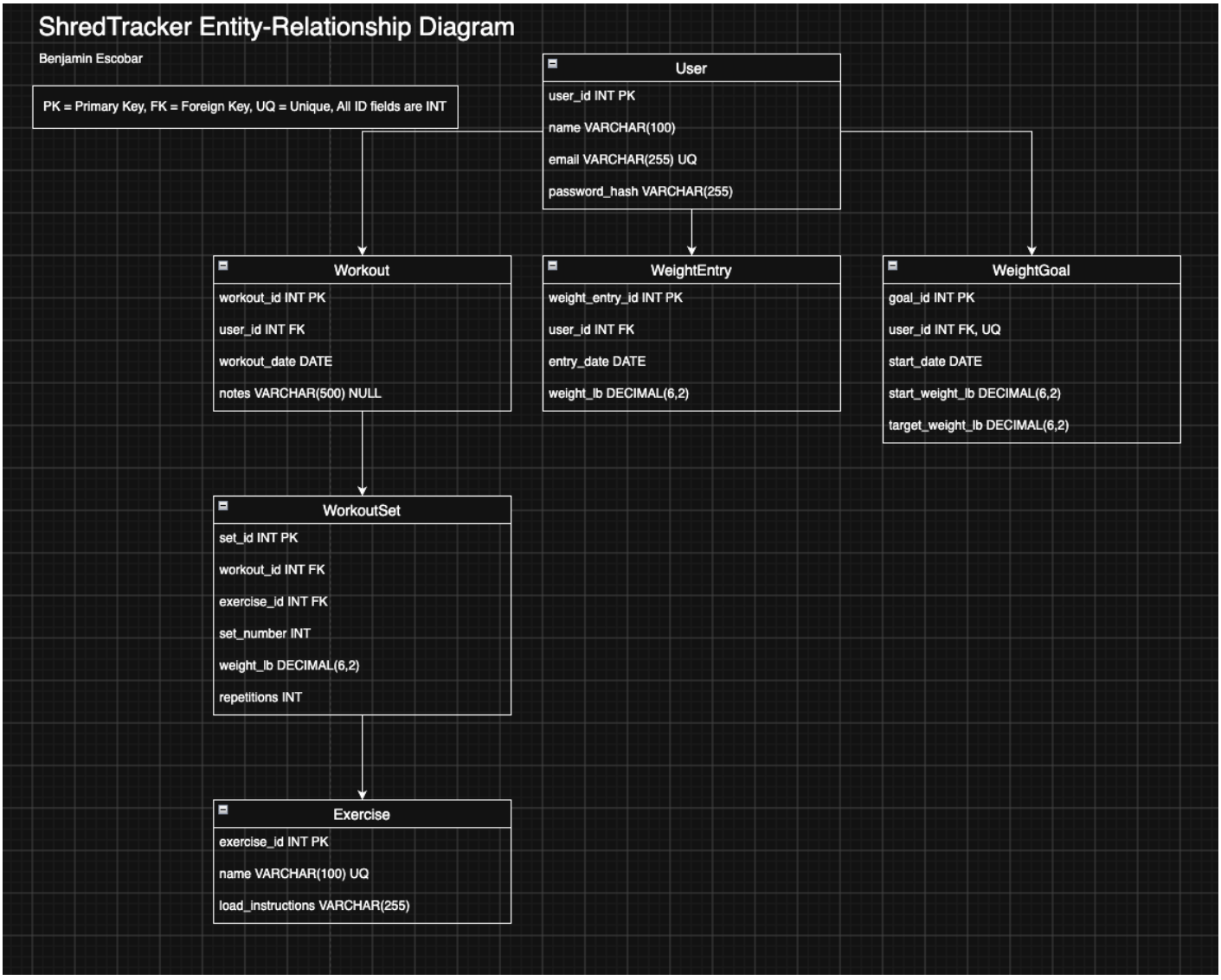
Architecture, Deployment and Design Scope

For its architecture it will use React for its interface to send requests to a Node.js/Express server layer. The service will validate entries, check which user owns the data, perform calculations and read or write MySQL records.

For deployment, Heroku will be the application host. Database credentials will be stored in a server configuration outside the source code.

For its design scope, MVP will use pounds for lifting loads and body weight. It will support one body weight entry per user a day, shared exercise list, multiple sets per workout and one user editable weight loss goal.

Entity-Relationship Diagram



Tables and Relationships

User

Will store each registered person and a password hash used for signing in. A user can have multiple workouts and entries. All ID primary keys are generated automatically and every field is required.

Workout and WorkoutSet

Workouts will store the owner, workout data and notes. WorkoutSet will store its workout, order, lifting weight and repetition, one row per set. Each workout will have zero or more sets while being entered, a completed workout will have at least one set. Each set will belong to one workout and one exercise. The workout_id, exercise_id and set_number are unique, so each exercise has its own numbered sets within that workout.

Exercise

This will store shared list of exercises and instructions for entering the lift load. These will use unique names. Workouts and Exercises will have many relationships through WorkoutSets. Since an exercise can appear in several workouts, keeping exercise details in one table will help be consistent with names in performance comparisons.

WeightEntry

This will store body weight entries with their user and calendar date. User_id and entry_date are unique. A user can save their weight in lbs. for one date and later correct the entry. The saved reading will supply the chart and current weight calculations.

WeightGoal

This will store the starting date, start weight and target weight for a user. User_id will be both a foreign key and unique. A user can start at goal with a target goal, but editing the goal will replace its value and the MVP will not keep the previous goals.

Data Validation and Database Access

Validation and Calculations

The repetition numbers must be whole and positive. Body weight must be positive and lifting loads cannot be negative. Target weight must be below the starting weight. Foreign

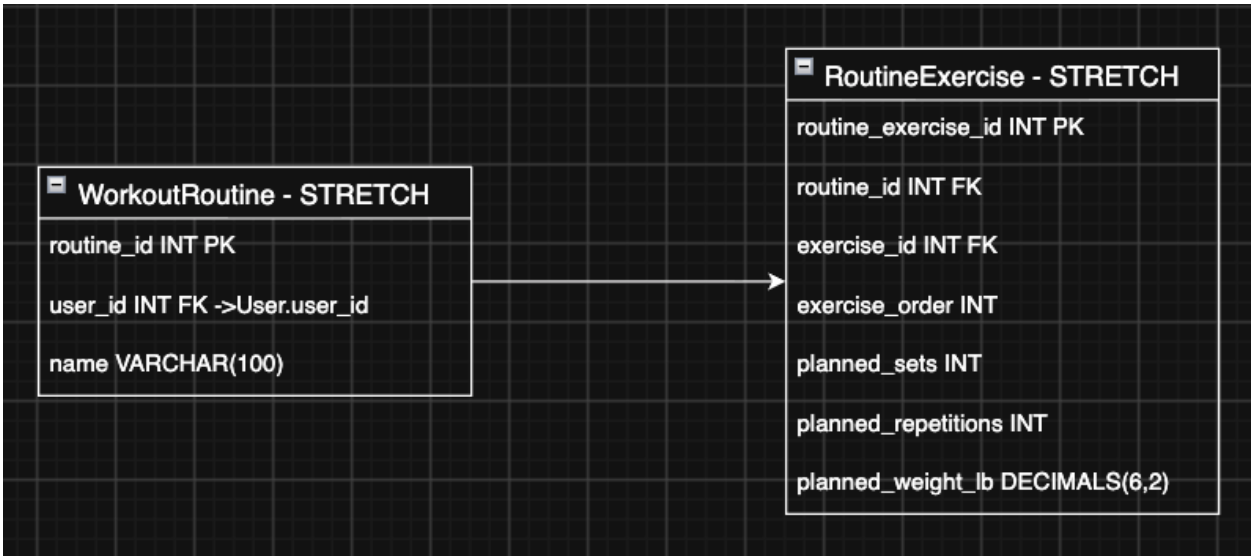
keys will connect related records. Goal progress and workouts will be calculated from saved data with editing and deleting records updating the results.

Database Access

To manage tables, database users, and exercise list, there will be an administrator account. There will also be an application account to read and update user's fitness records and exercise list. Users will also have access to their own records in the application. Passwords are stored as hashes.

Stretch Feature: Saved Routines

A user will be able to create a workout routine, save its planned exercises, and load it into the workout form. The user can adjust the values to what they completed before saving a Workout and its WorkoutSet records.



WorkoutRoutine

This will store the owner and name of a reusable plan. One User can have zero or many routines. Each routine will have only one owner.

RoutineExercise

This will store one exercise and its position and planned values in a routine. One exercise can appear in many routine entries.

Rules and relationship details

A saved routine must have at least one entry that will be enforced by API. The database relationship will permit zero or many entries during creation. Deleting a routine removes the routine entries. Deleing a user also removes their entries. Exercise deletion is blocked when referenced by another workout set or routine entry, and saving or updating a routine and its entries will happen together in one transaction. Loading a routine will copy its plan into form. Routine edits or deletions will not change previously save routines, and the UI and service will use this same stretch feature.